Predicting Parking Violation Rates

Using Gradient Boosting

ChallengeData #163 · CS5483 Data Mining · April 2026

Platform Spearman: 0.5705 · Rank #5 Globally

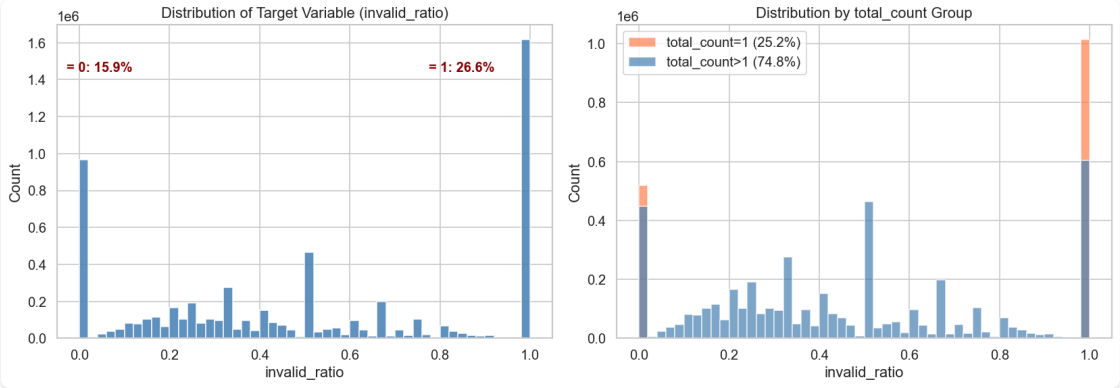
Section 1

Introduction & Problem Setup

What Are We Predicting?

- **Dataset**: THESi smart parking system, Thessaloniki, Greece
- **Training set**: 6.07 M observations, 10 features
- **Target**: `invalid_ratio` — fraction of invalid parking events per location-timeslot
- **Evaluation**: Spearman rank correlation ρ
- **Train / Test / Features**: 6.07 M · 1.5 M · 10

> **The metric rewards ranking, not numerical accuracy.**



Distribution of `invalid_ratio` — heavy mass at 0 and 1 from low-count noise

Understanding the Evaluation Metric

What is Spearman ρ?

- Measures **rank agreement** between predictions and ground truth
- Only **relative ordering** matters — not absolute values
- A perfect ranking scores ρ = 1.0 regardless of scale

	Objective	Cares About
❌ MSE	Minimize squared error	Absolute values
✅ Spearman ρ	Preserve rank order	Relative ordering

> *"Getting the order right matters more than getting the number right."*

How well did we do?

0.5705

Platform Spearman · Rank #5 Globally

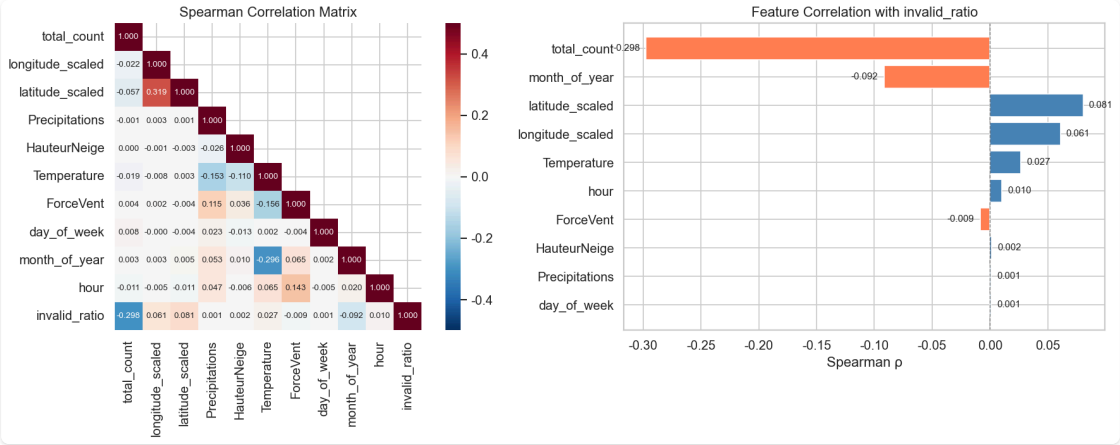
Benchmark	Score
Official baseline (RF, 10 trees)	0.197
Our final result	**0.5705**
Relative improvement	**+190%**

Section 2

Data Exploration & Feature Engineering

What the Data Tells Us

- **Strongest predictor**: `total_count` (ρ = −0.297) — busier locations have fewer violations
- Geographic patterns carry strong spatial signal
- Temporal features show enforcement cycles
- Weather features: minimal predictive power (ρ < 0.03)



Spearman correlations — total_count and location dominate

Challenge: High Noise in Low-Count Observations

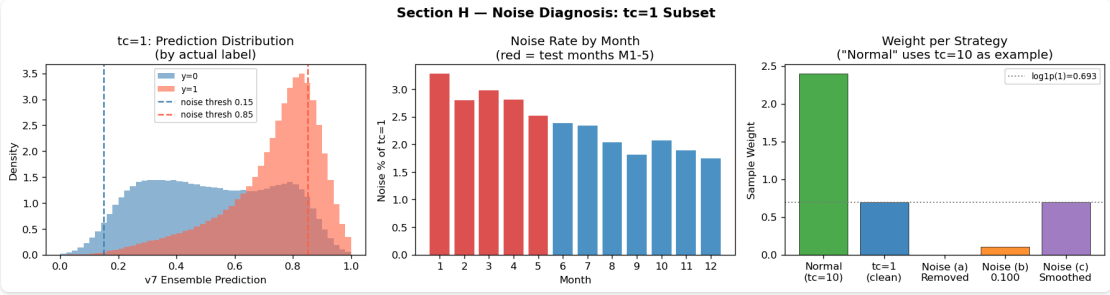
- Locations with `total_count = 1` account for **~25% of training data**
- For these, `invalid_ratio` is exactly 0 or 1 — binary, not continuous
- Creates severe **label noise** that degrades model training

Solution: Sample Weighting

```
sample_weight = np.log1p(total_count)
```

- Downweights noisy tc=1 samples without discarding them
- Preserves the full 6 M training set

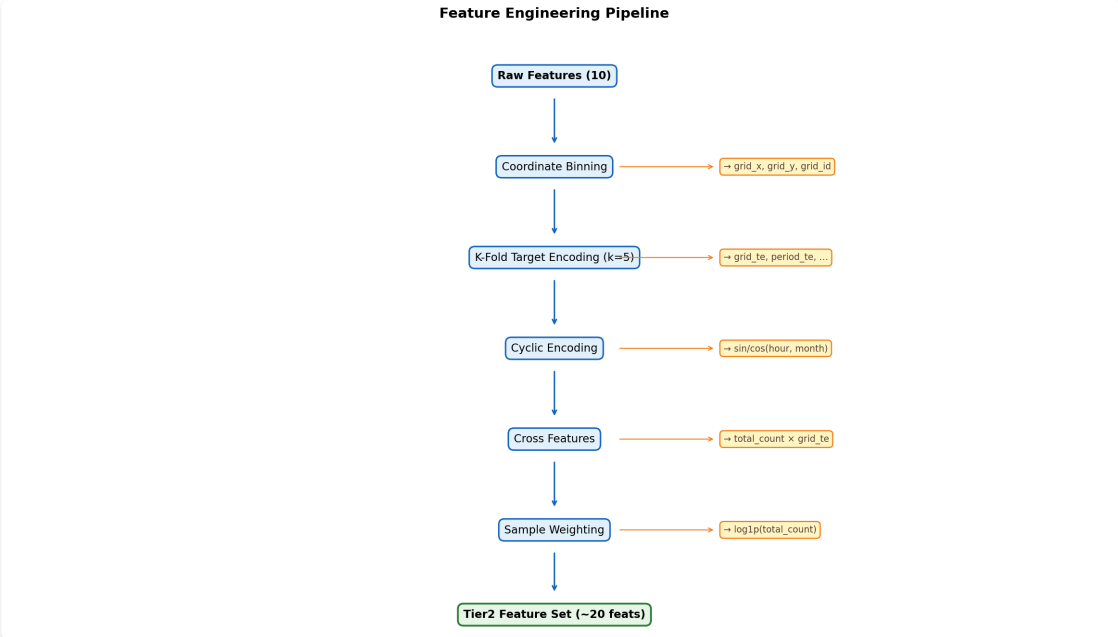
> *"Down-weight unreliable samples, don't throw them away."*



Label noise diagnosis — tc=1 subset shows extreme bimodality

Tier 2 Feature Engineering Pipeline

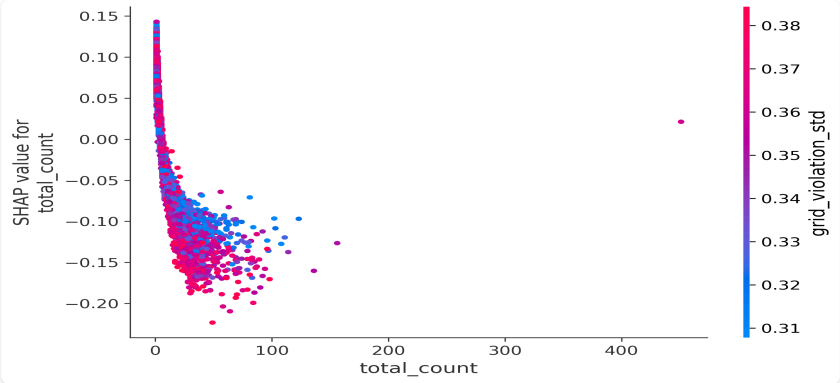
1. **Spatial binning**: Divide map into grids → `grid_x`, `grid_y`, `grid_id`
2. **K-Fold Target Encoding** (k=5): `grid_te`, `period_te` — captures violation rates per zone without leakage
3. **Cyclic encoding**: `sin/cos(hour)`, `sin/cos(month)` — preserves periodicity
4. **Cross features**: `total_count × grid_te` — busyness × location risk
5. **Result**: ~20 engineered features from 10 originals

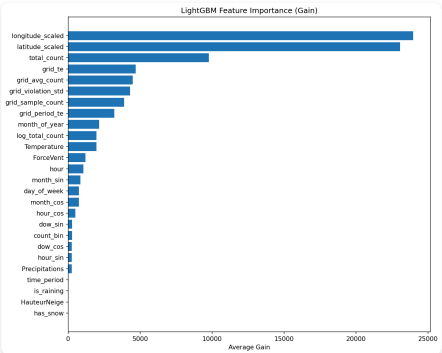


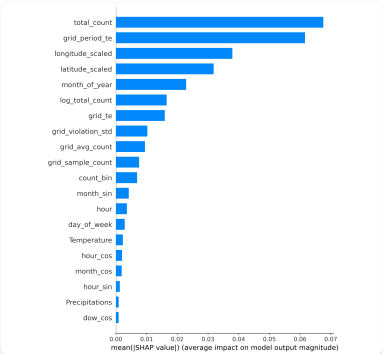
K-Fold TE prevents data leakage — a critical design choice.

What Drives Violation Rates?

- `total_count` and `grid_te` are consistently the **top-2 features**
- Geographic (TE) features dominate over weather and raw coordinates
- SHAP: high `total_count` → **lower violation rate**
- Weather: near-zero SHAP contribution







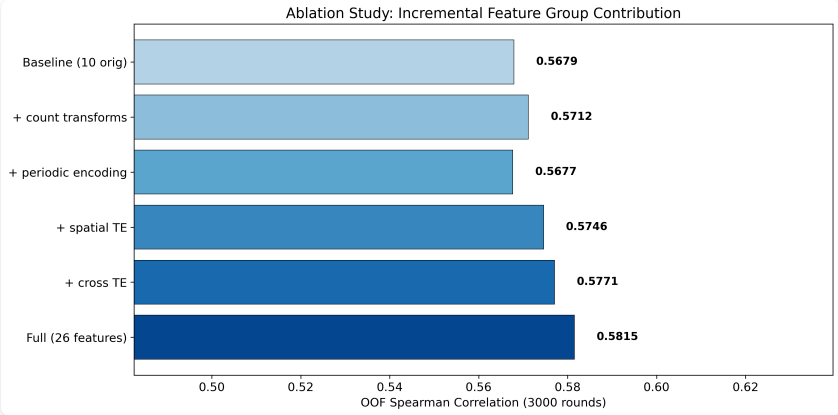
Section 3

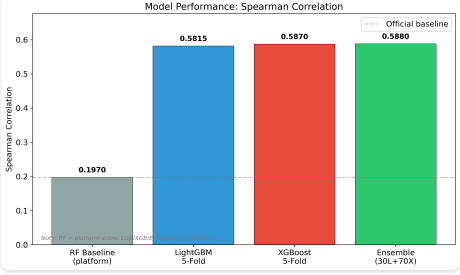
Baseline Development & Gap Analysis

Model: LightGBM + XGBoost Ensemble

- **Base models**: LightGBM and XGBoost (gradient boosting)
- **Ensemble**: weighted average, weights optimized on OOF Spearman
- CatBoost tested — final weight → **0**, excluded
- **Cross-validation**: 5-Fold with Spearman early stopping

Model	OOF ρ
LightGBM alone	~0.630
XGBoost alone	~0.618
LGB + XGB ensemble	**0.6429**
+ CatBoost	no gain





Iterative Improvement: v1 → v7

Version	Key Change	Platform ρ	Δ
v1	Initial LGB + XGB	0.5222	—
v2	Increased n_estimators	0.5338	+0.0116
v3	Optuna hyperparameter tuning	0.5620	+0.0282
v7	Sample weighting `log1p(tc)`	**0.5636**	+0.0016

- Optuna (v3) was the biggest pre-innovation gain: **+0.0282**
- OOF ≈ 0.643, Platform ≈ 0.564 — consistent gap ~0.079

> Each engineering decision produced **measurable, documented improvement.**

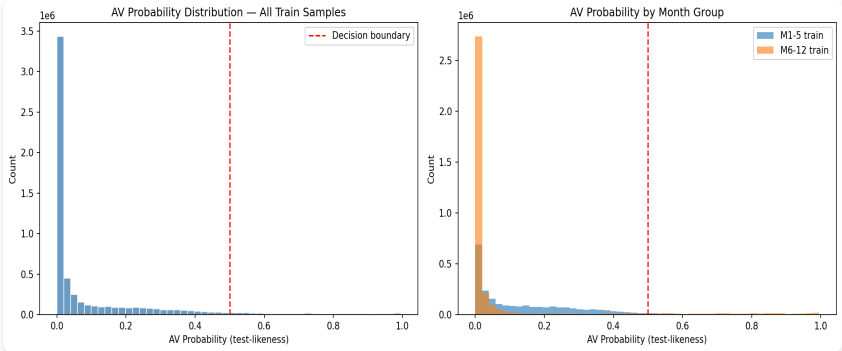
Why Is There a Gap Between OOF and Platform?

Observed: OOF ~0.643, Platform ~0.564 → gap **~0.079**

Hypothesis	Test	Result
Overfitting	Stronger regularization	❌ Worse scores
Distribution shift	Adversarial validation	✅ AUC = 0.9999
	Temporal CV (M1–M4 → M5)	✅ Gap −0.041
	TE distribution plot	✅ Clear mismatch

Train and test come from **different temporal periods** — shift is structural.

> *"Diagnose before you optimize."*



Top: near-perfect AV separation. Bottom: stable ~0.077 gap.

Section 4

Key Innovation: Rank-Target Training

Training with MSE ≠ Optimizing Spearman

MSE penalizes large **numerical** deviations.
Spearman only cares about **relative order**.

Training in the wrong direction.

Solution: Train to Rank, Not to Regress

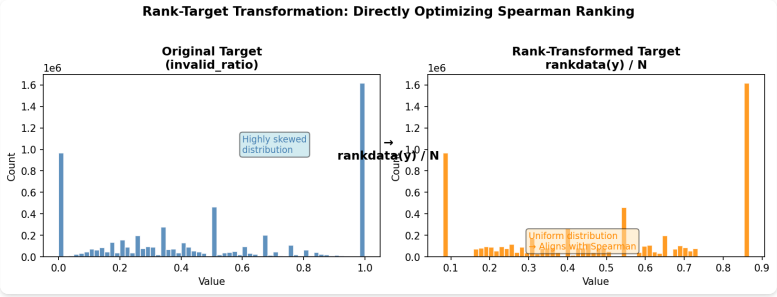
Transformation:

$$y_{\text{rank}} = \frac{\text{rankdata}(y)}{N}$$

- Converts target to **uniform [0, 1] distribution** of ranks
- Model learns **relative ordering**, not absolute values
- No other changes to the pipeline

```
# Replace raw target with rank-normalized target
y_rank = rankdata(train_df['invalid_ratio']) / len(train_df)
# Train exactly as before
lgb_model.fit(X_train, y_rank, sample_weight=weights, ...)
```

> *"One line of code. The biggest single improvement in our entire pipeline."*



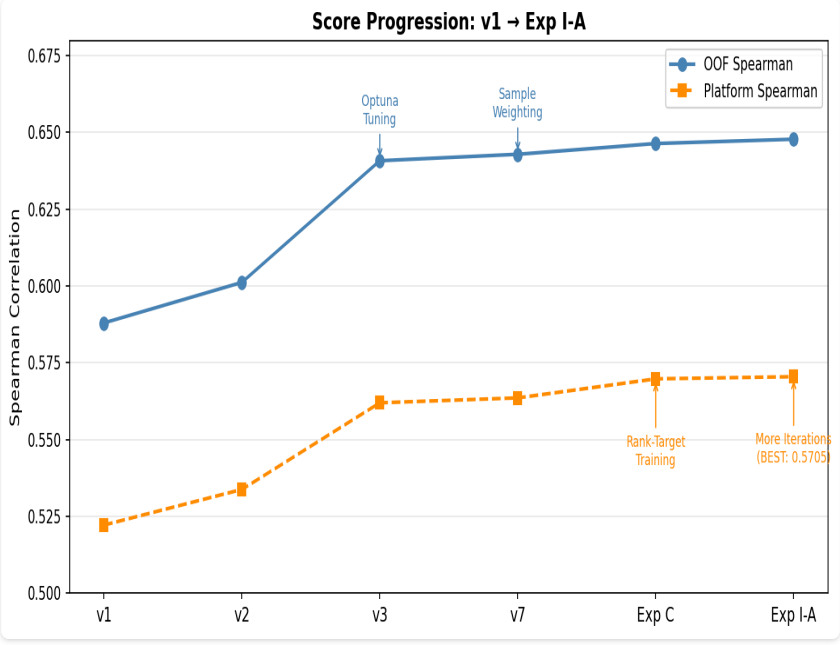
Left: skewed original target. Right: uniform rank-transformed target.

Impact: Rank-Target Delivers Our Largest Gain

	OOF ρ	Platform ρ	Δ Platform
v7 (baseline)	0.6429	0.5636	—
Exp C (rank-target)	0.6464	0.5698	**+0.0062**
Exp I-A (+iterations)	**0.6478**	**0.5705**	**+0.0069**

- Exp C alone: rank-target gives **+0.0062** — more than any FE step
- Exp I-A: LGB iterations 10K → 20K → extra **+0.0007**
- Final: **Platform 0.5705, Rank #5 globally**

> Metric alignment = the **biggest single-step improvement** of the project.

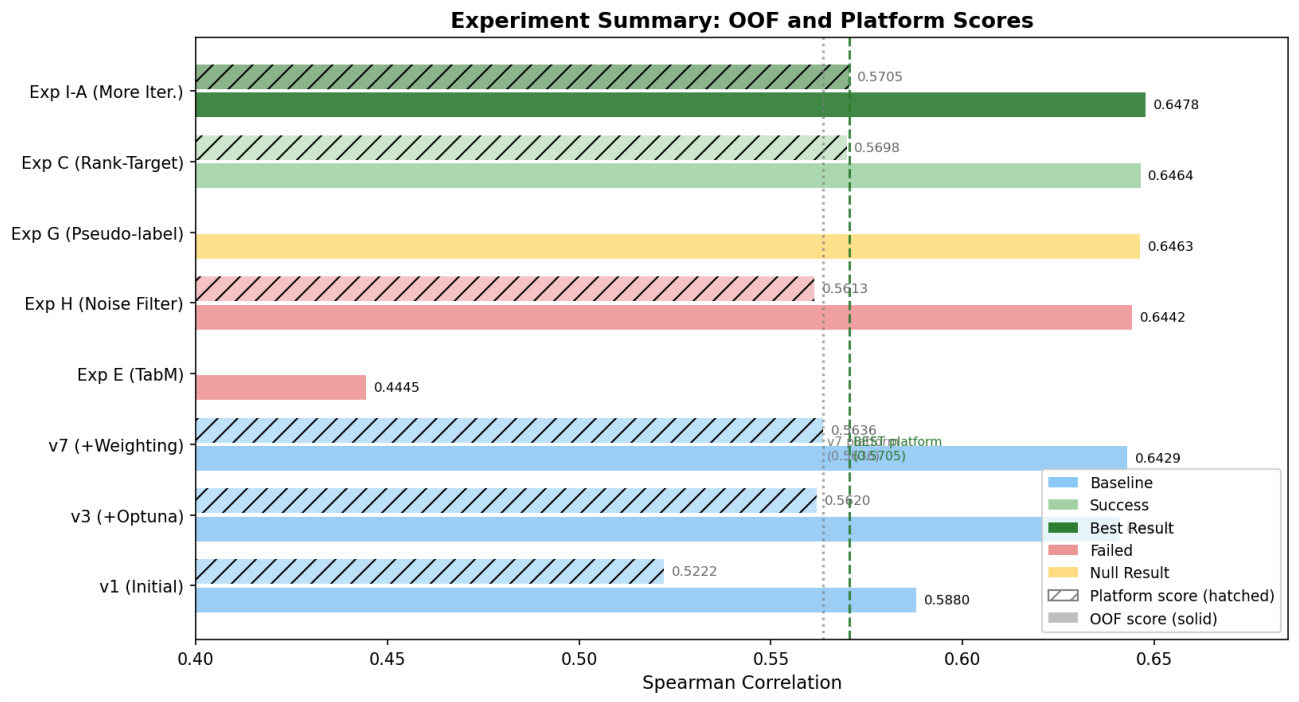


Full journey: v1 (0.5222) → Exp I-A (0.5705)

Section 5

Experiment Summary & Analysis

9 Experiments: What Worked and What Did Not



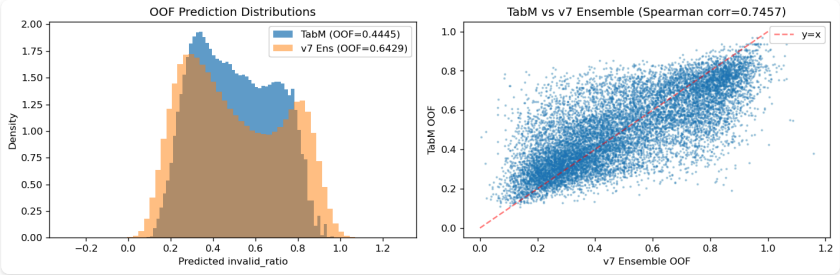
OOF (solid bars) · Platform (hatched bars) · **Green = success** · **Red = failed** · **Yellow = null result**

Deep Learning Does Not Help Here

- Tested **TabM** (ICLR 2025 — state-of-the-art tabular DL)
- OOF Spearman: **0.4445** vs GBDT **0.6429** — gap of **0.198**
- Root cause: only 10 input features, no image/text structure → GBDT advantages dominate on compact tabular data

Model	OOF ρ	Notes
TabM (ICLR 2025)	0.4445	Best DL attempt
LGB + XGB (v7)	0.6429	GBDT baseline
Rank-Target (Exp I-A)	**0.6478**	Our final

> *"Domain structure matters more than model architecture."*

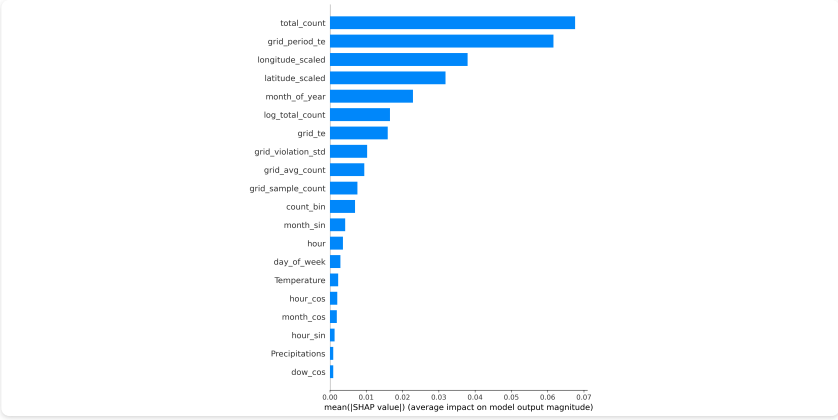


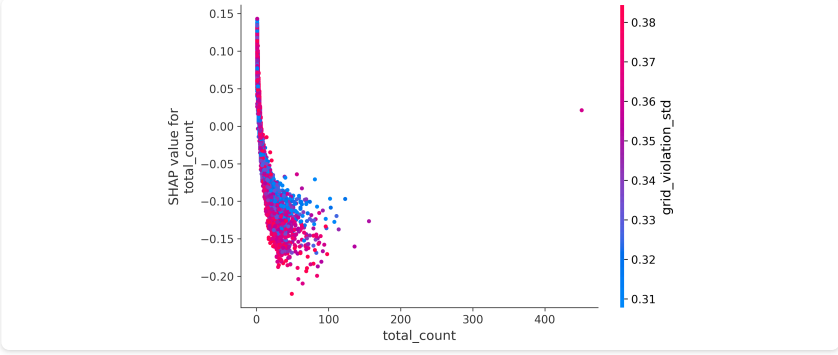
TabM predictions poorly correlated with ground truth

Understanding the Model with SHAP

- `total_count` : strong **negative** SHAP — busy locations comply more
- `grid_te` : captures spatial violation risk per zone
- Weather: near-zero SHAP — no meaningful contribution
- Model decisions are **explainable** and match domain intuition

Insight: the model learns *"where are the risky zones, and how busy are they right now?"*





Section 6
Conclusion

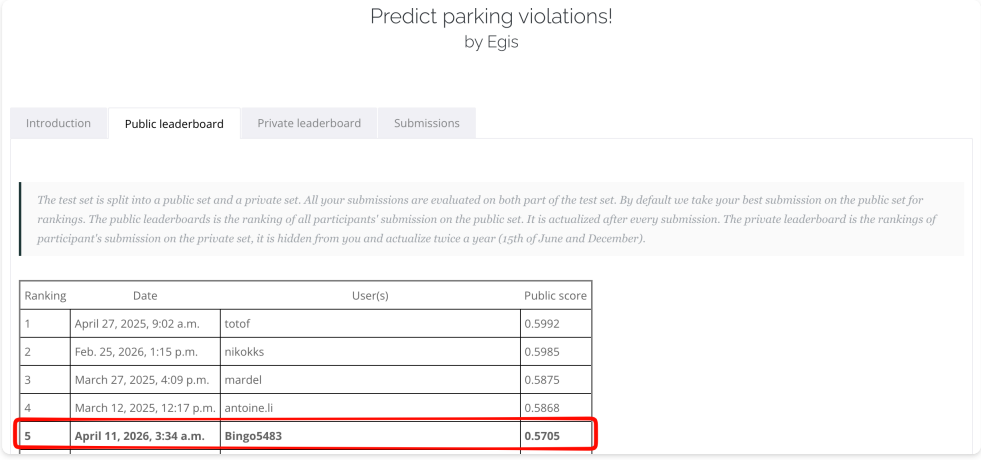
Results: Platform 0.5705, Rank #5

- Official baseline: **0.197** → Our result: **0.5705** (+190%)
- Leaderboard: **Rank #5 globally**

Key contributions:

1. **Tier 2 feature engineering** — leakage-free K-Fold TE
2. **Rank-target training** — aligning objective with Spearman
3. **Systematic gap diagnosis** via adversarial validation
4. **Sample weighting** to handle label noise

🔗 Public Leaderboard



Predict parking violations!
by Egis

Introduction | Public leaderboard | Private leaderboard | Submissions

The test set is split into a public set and a private set. All your submissions are evaluated on both part of the test set. By default we take your best submission on the public set for rankings. The public leaderboards is the ranking of all participants' submission on the public set. It is actualized after every submission. The private leaderboard is the rankings of participant's submission on the private set, it is hidden from you and actualize twice a year (15th of June and December).

Ranking	Date	User(s)	Public score
1	April 27, 2025, 9:02 a.m.	totof	0.5992
2	Feb. 25, 2026, 1:15 p.m.	nikokks	0.5985
3	March 27, 2025, 4:09 p.m.	mardel	0.5875
4	March 12, 2025, 12:17 p.m.	antoine.li	0.5868
5	**April 11, 2026, 3:34 a.m.**	**Bingo5483**	**0.5705**

Rank #5 globally with Platform Spearman 0.5705

Key Takeaways

Lesson 1: Match your training objective to your evaluation metric

→ Rank-target training was the single biggest improvement in the project (+0.0069 Platform Spearman)

Lesson 2: Diagnose before you optimize

→ The OOF-Platform gap was structural distribution shift — stronger regularization would have made it worse

Lesson 3: Systematic iteration with quantitative baselines outperforms guesswork

→ Every version tracked, every change measured, every null result documented

Future directions: ensemble stacking · richer spatial features (POI density, road type) · cross-city generalization